

Arithmetic Dyadic Rays of the Rvachev Fourier Product

Cyclotomic orbit norms, exact periodic decay, and a half-base q -binomial bridge

Frontier research report based on the complete `Analysis/FabiusFunction/docs` corpus

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Abstract

The documentation corpus under `Analysis/FabiusFunction/docs` contains a mature, Lean-backed exposition of the Fabius function and Rvachev's compactly supported up-function, a large quarantine volume of non-formalized results, a Thue–Morse formula atlas, twelve competing and synthesized studies of the Fourier decay of up, seven source-paper transcriptions, and thirty-two frozen historical TeX revisions. This report audits all fifty-seven TeX corpus items, distinguishes results already present from genuinely unfilled directions, and develops an arithmetic theory for

$$\mathcal{U}(x) := \widehat{\text{up}}(x) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^n}\right).$$

The central result is an exact decay law on every non-dyadic rational ray $x_N = 2^N(1 + a/q)$. If q is odd, $d = \text{ord}_q(2)$, and

$$A_{a,q} = \left| \prod_{j=0}^{d-1} (1 - \zeta_q^{a2^j}) \right|,$$

then

$$\log |\mathcal{U}(x_N)| = -\frac{(\log x_N)^2}{2 \log 2} - \kappa_{a,q} \log x_N + \Omega_{a,q}(N), \quad \kappa_{a,q} = \frac{1}{2} + \frac{\log(2\pi)}{\log 2} - \frac{\log A_{a,q}}{d \log 2},$$

where $\Omega_{a,q}$ is explicitly periodic in N . The multiplier is a relative cyclotomic norm of a Thue–Morse polynomial value. We construct its integral orbit polynomial by a resultant, express its power traces through Thue–Morse signs and Ramanujan sums, and prove an exact mean-exponent law over Galois orbits. A short periodic-orbit argument gives the sharp bound $A_{a,q}^{1/d} \leq \sqrt{3}$, with equality only for the orbit $\{1/3, 2/3\}$.

For the denominators $2^d - 1$ and $2^d + 1$ we obtain a refined two-sided expansion. Both exponents grow as $d/2 + 1 - (\log h)/(d \log 2)$, where $h = \prod_{k \geq 1} \text{sinc}(\pi/2^k) = -\mathcal{U}'(1)$, while the first exponentially small corrections have opposite signs. Combining this with $\mathfrak{M}_d = \prod_{j=1}^d (2^j - 1) = 2^{d(d+1)/2} (1/2; 1/2)_d$ yields a direct limit identity joining the half-base q -Pochhammer normalization, a cyclotomic orbit norm, and the Rvachev sinc tail. All statements marked “new” are proved below. Novelty is claimed relative to the reviewed repository corpus and the targeted literature search; the root-of-unity recurrence itself is explicitly credited to Sobolewski's broader automatic-polynomial theory.

Proof-status convention. `IMPORTED RESULT`. means that a result is already in the repository or cited literature. `NEW RESULT PROVED HERE`. means that a complete proof is supplied here and the statement was not found in the reviewed corpus or targeted literature search. `OPEN OR CONJECTURAL`. labels questions and numerical conjectures. This convention prevents known root-of-unity identities from being relabeled as discoveries merely because they acquire a new application to the Rvachev product.

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1 Executive summary and frontier boundary

1.1 The contribution map

Result	Content	Status	
Rational-ray factorization	Splits $\mathcal{U}(2^N y)$ into a universal log-Gaussian factor, a smooth mantissa tail, and a finite Thue–Morse product.	NEW	RESULT PROVED HERE.
Exact periodic ray law	Gives $\kappa_{a,q}$ and an explicit period- d correction $\Omega_{a,q}((N + 1) \bmod d)$, with no error term.	NEW	RESULT PROVED HERE.
Cyclotomic orbit polynomial	Constructs $\mathcal{P}_{q,2}(X) \in \mathbb{Z}[X]$ and proves a resultant identity whose right side is a perfect d th power.	NEW	RESULT PROVED HERE.
Ramanujan trace bridge	Expresses all power traces of the orbit multipliers as finite Thue–Morse–Ramanujan sums.	NEW	RESULT PROVED HERE.
Mean exponent law	Computes the Galois-orbit mean of $\kappa_{a,q}$ from the single integer $\Phi_q(1)$.	NEW	RESULT PROVED HERE.
Sharp endpoint	Proves $A_{a,q}^{1/d} \leq \sqrt{3}$, uniquely extremized by $q = 3$, and obtains the exact minimum rational-ray exponent.	NEW	RESULT PROVED HERE.
Mersenne–Fermat twin expansion	Gives matching main terms and opposite first corrections for denominators $2^d \mp 1$.	NEW	RESULT PROVED HERE.
q -binomial bridge constant	Shows that two Mersenne-normalized cyclotomic norms converge to $(1/2; 1/2)_\infty h$.	NEW	RESULT PROVED HERE.
Root-of-unity multiplier	The recurrence $T_{n+d}(\zeta) = \Gamma T_n(\zeta)$ and fixed-field interpretation.	IMPORTED	RESULT. [7]
Metric cocycle theory	Exact variance, central limit theorem, law of the iterated logarithm, transfer operator, and multifractal discussion.	IMPORTED	RESULT. from the Fourier-decay drafts

Table 1: The result boundary used throughout the report.

1.2 Why arithmetic rays are the genuine gap

The Fourier-decay subcorpus already goes far beyond the elementary statement that $\widehat{\text{up}}$ decreases faster than every power. Across its twelve drafts it develops the logarithmic cocycle for the doubling map, almost-everywhere behavior, exact variance, central-limit and iterated-logarithm fluctuations, extremal envelopes, transfer operators, and a moment spectrum. Re-deriving that material would add polish rather than frontier mathematics. The Thue–Morse atlas likewise contains

$$T_n(z) = \sum_{m=0}^{2^n-1} (-1)^{s_2(m)} z^m = \prod_{j=0}^{n-1} (1 - z^{2^j}),$$

its Mahler recursion, rarefied sums, roots-of-unity methods, cyclotomic factors, and extensive arithmetic subsequence material.

The missing composition was

Rvachev sinc product $\longleftrightarrow$ doubling-orbit sine product $\longleftrightarrow$ cyclotomic Thue–Morse norm.

For irrational mantissas the middle term is a nontrivial ergodic cocycle. For a rational mantissa it is periodic, and periodicity promotes an asymptotic heuristic to an exact finite-state law. That promotion is the organizing idea of this report.

1.3 Corpus and novelty audit

The review treated every TeX path listed in Appendix A as a corpus item. The live primary exposition, frontier volume, formula atlas, Fourier-decay studies, and source-paper transcriptions were reviewed

at source level. The thirty-two frozen archive items were compared as revision chains, so historical duplicates were not counted as independent mathematical claims. The archive's provenance notes were used to identify material unique to older revisions.

The external search concentrated on Fabius/Rvachev Fourier products, lacunary trigonometric products, Thue–Morse polynomials at roots of unity, cyclotomic properties of automatic polynomials, and exact dyadic arithmetic. Sobolewski's root-of-unity recurrence is the closest precursor: it supplies the multiplier and fixed-field structure, but the located work does not state the Rvachev rational-ray law, the orbit resultant in the form used below, the exponent mean law, the sharp $\sqrt{3}$ endpoint, or the Mersenne–Fermat/ q -Pochhammer bridge. The novelty claim is therefore deliberately limited: these deductions are new relative to the reviewed corpus and were not found explicitly in the located literature.

2 Established platform from the corpus

2.1 Fabius, Rvachev, and the Fourier product

The primary exposition carefully separates the bounded Fabius distribution function F , the even compactly supported Rvachev function up, and the signed global continuation governed by Thue–Morse signs [1, 2, 3]. For this report only up and its Fourier transform are needed. In the repository normalization,

$$\mathcal{U}(x) := \widehat{\text{up}}(x) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^n}\right), \quad \text{sinc } u = \frac{\sin u}{u}, \quad \text{sinc } 0 = 1. \quad (1)$$

The product is entire and obeys

$$\mathcal{U}(x) = \text{sinc}(\pi x) \mathcal{U}(x/2). \quad (2)$$

Repeated use of the double-angle identity also yields

$$\mathcal{U}(x) = \prod_{m=1}^{\infty} \left(\cos \frac{\pi x}{2^m} \right)^m. \quad (3)$$

2.2 The finite Thue–Morse polynomial

Write $\varepsilon_m = (-1)^{s_2(m)}$, where $s_2(m)$ counts the binary ones in m . The block polynomial is

$$T_n(z) := \sum_{m=0}^{2^n-1} \varepsilon_m z^m = \prod_{j=0}^{n-1} (1 - z^{2^j}). \quad (4)$$

The factorization follows from independent binary digit choices and encodes Prouhet cancellation: $z = 1$ is a zero of order n , hence

$$\sum_{m < 2^n} \varepsilon_m m^r = 0 \quad (0 \leq r < n).$$

The bridge to the Fourier product is

$$|T_n(e^{2\pi i t})| = \prod_{j=0}^{n-1} |1 - e^{2\pi i 2^j t}| = \prod_{j=0}^{n-1} |2 \sin(\pi 2^j t)|. \quad (5)$$

2.3 The half-base q -calculus layer

The exact dyadic-value theory uses the base $q = 1/2$ Pochhammer symbol and Gaussian binomial coefficients. In the repository normalization,

$$(z; 1/2)_n = \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2} z^k. \quad (6)$$

With the Mersenne factorial

$$\mathfrak{M}_n := \prod_{j=1}^n (2^j - 1), \quad (7)$$

one has

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2} = \frac{\mathfrak{M}_n}{\mathfrak{M}_k \mathfrak{M}_{n-k} 2^{k(n-k)}}, \quad (1/2; 1/2)_n = 2^{-n(n+1)/2} \mathfrak{M}_n. \quad (8)$$

The first q here is an analytic base fixed at $1/2$; later q denotes an odd cyclotomic denominator. Section 8 gives a limit in which these two roles meet naturally rather than notationally.

2.4 What is imported and what is not

The twelve Fourier-decay manuscripts repeatedly derive a decomposition of $\log |\mathcal{U}(2^N y)|$ into a universal quadratic term and a Birkhoff sum for $x \mapsto 2x \pmod{1}$. Their final syntheses include the typical linear exponent, exact covariance calculations, a central limit theorem, a law of the iterated logarithm, transfer-operator formulas, and a multifractal moment spectrum. Those results are context, not new contributions here.

For a primitive q th root ζ_q and $d = \text{ord}_q(2)$, the sequence $T_n(\zeta_q^a)$ obeys a two-term recurrence of span d . This is an elementary special case of Sobolewski's general recurrence theory for automatic-sequence polynomials at roots of unity [7]. The same work places the multiplier in the fixed field generated by $\langle 2 \rangle$ and proves integrality properties. We therefore mark the recurrence itself as imported. What follows from inserting it into the exact Rvachev factorization is the new contribution.

3 Exact fixed-mantissa factorization

Fix

$$1 < y < 2, \quad t := y - 1 \in (0, 1), \quad x_N := 2^N y,$$

and define

$$H(y) := \prod_{r=1}^{\infty} \text{sinc}\left(\frac{\pi y}{2^r}\right), \quad Q_n(t) := \prod_{j=0}^{n-1} |2 \sin(\pi 2^j t)|, \quad Q_0(t) := 1. \quad (9)$$

The tail H is positive and real analytic on $(0, 2)$. By (5),

$$Q_n(t) = |T_n(e^{2\pi i t})|. \quad (10)$$

Thus all ray-dependent arithmetic is concentrated in a finite Thue–Morse product.

Theorem 3.1 (Exact ray factorization). *For every $N \geq 0$ and $y \in (1, 2)$,*

$$\boxed{|\mathcal{U}(2^N y)| = 2^{-N(N+1)/2} (\pi y)^{-(N+1)} H(y) 2^{-(N+1)} Q_{N+1}(y-1).} \quad (11)$$

The identity remains valid when both sides vanish.

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Proof. Split the defining product at $n = N$ and put $k = N - n$ in the finite part:

$$\mathcal{U}(2^N y) = \prod_{k=0}^N \operatorname{sinc}(\pi 2^k y) \prod_{r=1}^{\infty} \operatorname{sinc}\left(\frac{\pi y}{2^r}\right).$$

The second product is $H(y)$. Since $y = 1 + t$ and 2^k is an integer, $|\sin(\pi 2^k y)| = |\sin(\pi 2^k t)|$. The denominator of the finite product is

$$\prod_{k=0}^N \pi 2^k y = (\pi y)^{N+1} 2^{N(N+1)/2},$$

whereas its numerator is $2^{-(N+1)} Q_{N+1}(t)$. Multiplication proves the claim. $\square$

Put

$$\lambda := \log 2, \quad L_N := \log x_N = N\lambda + \log y, \quad \kappa_0 := \frac{1}{2} + \frac{\log(2\pi)}{\lambda} = 3.1514961294723187980\dots, \quad (12)$$

and define

$$\Psi(y) := \log H(y) - \log(2\pi) - \frac{1}{2} \log y + \frac{\log(2\pi) \log y}{\lambda} + \frac{(\log y)^2}{2\lambda}. \quad (13)$$

Corollary 3.2 (Master cocycle identity). *If $Q_{N+1}(t) \neq 0$, then*

$$\log |\mathcal{U}(2^N y)| = -\frac{L_N^2}{2\lambda} - \kappa_0 L_N + \Psi(y) + \log Q_{N+1}(t). \quad (14)$$

If $Q_{N+1}(t) = 0$, both sides are interpreted as $-\infty$.

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Proof. Take logarithms in (11). The deterministic finite part is

$$-\frac{N(N+1)}{2} \lambda - (N+1) \log(2\pi y) + \log H(y).$$

Substituting $N = (L_N - \log y)/\lambda$ and collecting powers of L_N gives (14); the constant is exactly (13). $\square$

Remark 3.3 (Relation with the metric theory). For variable t ,

$$\log Q_n(t) = \sum_{j=0}^{n-1} \log |2 \sin(\pi 2^j t)|$$

is the Birkhoff sum studied in the Fourier-decay drafts. The present report fixes t rational. The doubling orbit then closes, turning the Birkhoff sum into a finite algebraic cycle.

4 Exact arithmetic laws on rational dyadic rays

4.1 The cycle multiplier

Let $q > 1$ be odd, $(a, q) = 1$, and write

$$d := \operatorname{ord}_q(2), \quad \zeta_q := e^{2\pi i/q}, \quad H_q := \langle 2 \rangle \subseteq (\mathbb{Z}/q\mathbb{Z})^\times. \quad (15)$$

Define

$$\Gamma_{a,q} := \prod_{j=0}^{d-1} (1 - \zeta_q^{a2^j}) = T_d(\zeta_q^a), \quad A_{a,q} := |\Gamma_{a,q}| > 0, \quad \alpha_{a,q} := \frac{1}{d} \log A_{a,q}. \quad (16)$$

The value depends only on the coset aH_q .

Proposition 4.1 (Root-of-unity block recurrence). *For every $n \geq 0$,*

$$T_{n+d}(\zeta_q^a) = \Gamma_{a,q} T_n(\zeta_q^a). \quad (17)$$

Consequently, if $n = md + r$, $0 \leq r < d$, then

$$Q_n(a/q) = A_{a,q}^m Q_r(a/q). \quad (18)$$

IMPORTED RESULT. in its general automatic-polynomial setting; the special-case proof is included.

Proof. Using (4) and $2^d \equiv 1 \pmod{q}$,

$$\begin{aligned} T_{n+d}(\zeta_q^a) &= \prod_{j=0}^{n+d-1} (1 - \zeta_q^{a2^j}) \\ &= \left(\prod_{j=0}^{d-1} (1 - \zeta_q^{a2^j}) \right) \left(\prod_{j=d}^{n+d-1} (1 - \zeta_q^{a2^j}) \right) \\ &= \Gamma_{a,q} \prod_{j=0}^{n-1} (1 - \zeta_q^{a2^j}). \end{aligned}$$

Iteration and absolute values give (18). □

4.2 The exact log-quadratic law

Set

$$y_{a,q} := 1 + \frac{a}{q} \in (1, 2), \quad x_N := 2^N y_{a,q}, \quad (19)$$

and define the arithmetic ray exponent

$$\boxed{\kappa_{a,q} := \kappa_0 - \frac{\alpha_{a,q}}{\lambda} = \frac{1}{2} + \frac{\log(2\pi)}{\log 2} - \frac{\log A_{a,q}}{d \log 2}.} \quad (20)$$

For $0 \leq r < d$, put

$$\Omega_{a,q}(r) := \Psi(y_{a,q}) + \left(1 - \frac{\log y_{a,q}}{\lambda}\right) \alpha_{a,q} + \log Q_r(a/q) - r \alpha_{a,q}. \quad (21)$$

Theorem 4.2 (Exact rational-ray decay law). *For every $N \geq 0$, let r_N be the residue of $N + 1$ modulo d in $\{0, \dots, d - 1\}$. Then*

$$\boxed{\log |\mathcal{U}(2^N y_{a,q})| = -\frac{(\log x_N)^2}{2 \log 2} - \kappa_{a,q} \log x_N + \Omega_{a,q}(r_N).} \quad (22)$$

The complete remainder is periodic in N with period dividing d ; there is no asymptotic error term.

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Proof. Write $n = N + 1 = md + r_N$. Proposition 4.1 gives

$$\log Q_{N+1}(a/q) = (N + 1) \alpha_{a,q} + \log Q_{r_N}(a/q) - r_N \alpha_{a,q}.$$

Since $N + 1 = (L_N - \log y_{a,q})/\lambda + 1$, substitution into (14) changes the coefficient of L_N from $-\kappa_0$ to $-\kappa_{a,q}$ and leaves (21). □

Corollary 4.3 (Exact d -step recurrence along a ray). *Under the hypotheses above,*

$$|\mathcal{U}(2^{N+d}y_{a,q})| = A_{a,q} 2^{-d(d+3)/2} \pi^{-d} (2^N y_{a,q})^{-d} |\mathcal{U}(2^N y_{a,q})|. \quad (23)$$

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Proof. Take the quotient of (11) at $N + d$ and N . The tail cancels and $Q_{N+d+1}/Q_{N+1} = A_{a,q}$; simplifying the remaining powers gives the stated factor. $\square$

Remark 4.4 (Discrete scale invariance). After removal of the universal log-Gaussian and linear factors, the values on a rational ray lie on a finite cycle. The scale is 2^d , and the multiplier is not a fitted phenomenological constant but the cyclotomic norm (16).

4.3 All rational mantissas

Every reduced rational $t \in (0, 1)$ has a unique representation

$$t = \frac{a}{2^e q}, \quad e \geq 0, \quad q \text{ odd}, \quad (a, 2q) = 1. \quad (24)$$

For $n > e$,

$$Q_n(t) = Q_e(t) Q_{n-e}(a/q). \quad (25)$$

Corollary 4.5 (Zero-or-periodic dichotomy). *Let t be as in (24).*

1. *If $q = 1$, then $Q_n(t) = 0$ for every $n > e$, and $\mathcal{U}(2^N(1+t)) = 0$ for every $N \geq e$.*
2. *If $q > 1$, the finite prefix $Q_e(t)$ is followed by the exact law of [theorem 4.2](#), with period $\text{ord}_q(2)$.*

Thus a rational ray is either eventually an exact zero ray or an eventually periodic nonzero ray.

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Proof. Split the product defining Q_n at $j = e$. If $q = 1$, the factor at $j = e$ is $2|\sin(\pi a)| = 0$. If $q > 1$, the subsequent reduced odd-denominator orbit never hits zero and (18) applies. $\square$

Example 4.6 (The denominator-three ray). For $q = 3$, $d = 2$ and

$$\Gamma_{1,3} = (1 - \zeta_3)(1 - \zeta_3^2) = 3, \quad A_{1,3} = 3.$$

Hence

$$\kappa_{1,3} = \frac{1}{2} + \frac{\log(2\pi/\sqrt{3})}{\log 2} = 2.3590148791117407073 \dots$$

The periodic correction has at most two values.

Example 4.7 (A neutral cycle at $q = 15$). The subgroup $\langle 2 \rangle$ has order 4. Both cosets have multiplier -1 , so $A_{a,15} = 1$ and the linear exponent is exactly κ_0 . The orbit polynomial is $(X + 1)^2$, showing that it can have repeated roots when the multiplier lies in a smaller subfield.

Example 4.8 (The first visible split at $q = 17$). Here $d = 8$ and the multiplier polynomial is

$$X^2 - 34X + 17.$$

Its roots are $17 \pm 4\sqrt{17}$, with amplitudes $0.50757749752936 \dots$ and $33.492422502471 \dots$. The corresponding ray exponents are $3.2737836274 \dots$ and $2.5182757764 \dots$.

5 Cyclotomic algebra of the ray spectrum

5.1 Relative norms and orbit polynomials

Let $K_q = \mathbb{Q}(\zeta_q)$, $G_q = (\mathbb{Z}/q\mathbb{Z})^\times$, and let $K_q^{H_q}$ be the fixed field of $H_q = \langle 2 \rangle$. Then

$$\Gamma_{a,q} = N_{K_q/K_q^{H_q}}(1 - \zeta_q^a). \quad (26)$$

This imported fixed-field interpretation explains that every multiplier is an algebraic integer. The quotient G_q/H_q indexes its conjugates, with possible repetitions. Write

$$c_q^* := |G_q/H_q| = \frac{\varphi(q)}{d}. \quad (27)$$

Definition 5.1 (Cyclotomic orbit polynomial). Choose a representative a_C from every coset $C \in G_q/H_q$ and set

$$\mathcal{P}_{q,2}(X) := \prod_{C \in G_q/H_q} (X - \Gamma_{a_C,q}). \quad (28)$$

Theorem 5.2 (Integrality and resultant construction). Let $F_d(z) := T_d(z) = \prod_{j=0}^{d-1} (1 - z^{2^j})$. Then

$$\mathcal{P}_{q,2}(X) \in \mathbb{Z}[X] \quad (29)$$

and

$$\boxed{\text{Res}_z(\Phi_q(z), X - F_d(z)) = \mathcal{P}_{q,2}(X)^d.} \quad (30)$$

The polynomial on the right need not be squarefree.

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Proof. The coefficients of (28) are symmetric polynomials in algebraic integers, hence algebraic integers. The Galois group permutes the cosets, so every coefficient lies in $\mathbb{Q}$ and therefore in $\mathbb{Z}$.

Since Φ_q is monic with roots ζ_q^a , $a \in G_q$,

$$\text{Res}_z(\Phi_q(z), X - F_d(z)) = \prod_{a \in G_q} (X - F_d(\zeta_q^a)).$$

The value $F_d(\zeta_q^a) = \Gamma_{a,q}$ is constant on a coset aH_q , and every coset contains d elements. Grouping factors proves (30). $\square$

5.2 Ramanujan traces

Recall the Ramanujan sum

$$c_q(m) := \sum_{a \in G_q} \zeta_q^{am} = \mu\left(\frac{q}{(q,m)}\right) \frac{\varphi(q)}{\varphi(q/(q,m))}. \quad (31)$$

Theorem 5.3 (Power traces as finite digital sums). For every integer $r \geq 1$,

$$\boxed{\sum_{C \in G_q/H_q} \Gamma_{a_C,q}^r = \frac{1}{d} \sum_{0 \leq m_1, \dots, m_r < 2^d} \varepsilon_{m_1} \cdots \varepsilon_{m_r} c_q(m_1 + \cdots + m_r).} \quad (32)$$

In particular,

$$\text{Tr}_{K_q^{H_q}/\mathbb{Q}}(\Gamma_{1,q}) = \frac{1}{d} \sum_{m=0}^{2^d-1} \varepsilon_m c_q(m), \quad (33)$$

with the natural embedding multiplicities if $\Gamma_{1,q}$ lies in a smaller subfield.

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Proof. Expand

$$\Gamma_{a,q} = T_d(\zeta_q^a) = \sum_{m=0}^{2^d-1} \varepsilon_m \zeta_q^{am}.$$

The value is constant on each H_q -coset, so summing over cosets is $1/d$ times summing over all $a \in G_q$. Raise the finite sum to the r th power, interchange sums, and apply (31). $\square$

Corollary 5.4 (Newton reconstruction). *The first c_q^* power sums in (32), together with Newton's identities, reconstruct $\mathcal{P}_{q,2}(X)$ using integer arithmetic only. This is an independent check on the resultant method.*

5.3 A one-integer law for the mean exponent

Theorem 5.5 (Norm, constant term, and mean ray exponent). *With $c_q^* = \varphi(q)/d$,*

$$\prod_{C \in G_q/H_q} \Gamma_{a_C,q} = \Phi_q(1), \quad (34)$$

$$\mathcal{P}_{q,2}(0) = (-1)^{c_q^*} \Phi_q(1), \quad (35)$$

$$\prod_{C \in G_q/H_q} A_{a_C,q} = \Phi_q(1), \quad (36)$$

and

$$\frac{1}{c_q^*} \sum_{C \in G_q/H_q} \kappa_{a_C,q} = \kappa_0 - \frac{\log \Phi_q(1)}{\varphi(q) \log 2}. \quad (37)$$

For $q > 1$,

$$\Phi_q(1) = \begin{cases} p, & q = p^s \text{ is a prime power,} \\ 1, & q \text{ has at least two distinct prime divisors.} \end{cases} \quad (38)$$

Thus every non-prime-power denominator has mean exponent exactly κ_0 .

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Proof. By transitivity of norms,

$$\prod_C \Gamma_{a_C,q} = N_{K_q^{H_q}/\mathbb{Q}} \left(N_{K_q/K_q^{H_q}}(1 - \zeta_q) \right) = N_{K_q/\mathbb{Q}}(1 - \zeta_q) = \Phi_q(1).$$

This proves (34); the constant term and absolute-value product follow. Taking logarithms in (36), inserting (20), and using $dc_q^* = \varphi(q)$ gives (37). Formula (38) is the classical value of a cyclotomic polynomial at one. $\square$

Remark 5.6 (Compensation between conjugate rays). For a denominator with at least two prime factors, a coset with $A_{a,q} > 1$ and hence a smaller-than- κ_0 exponent must be compensated geometrically by cosets with $A_{a,q} < 1$. The $q = 17$ and $q = 31$ tables below show this splitting explicitly.

6 The sharp rational-ray endpoint

Theorem 6.1 (Sharp periodic-orbit sine-product bound). *For every odd $q > 1$ and every $(a, q) = 1$,*

$$A_{a,q}^{1/d} \leq \sqrt{3}. \quad (39)$$

Equality holds if and only if $q = 3$ (with $a = 1$ or 2).

NEW RESULT PROVED HERE.

Proof. Let

$$t_j = \left\{ \frac{a2^j}{q} \right\}, \quad c_j = \cos(2\pi t_j), \quad 0 \leq j < d,$$

with indices cyclic. Doubling gives $c_{j+1} = 2c_j^2 - 1$. If $s = d^{-1} \sum_j c_j$, cyclic averaging and Cauchy-Schwarz give

$$s = 2 \frac{1}{d} \sum_j c_j^2 - 1 \geq 2s^2 - 1.$$

Hence $(2s + 1)(s - 1) \leq 0$ and $s \geq -1/2$. By arithmetic-geometric mean,

$$\begin{aligned} A_{a,q}^{2/d} &= \left(\prod_{j=0}^{d-1} |1 - e^{2\pi i t_j}|^2 \right)^{1/d} \\ &= \left(\prod_{j=0}^{d-1} (2 - 2c_j) \right)^{1/d} \leq \frac{1}{d} \sum_{j=0}^{d-1} (2 - 2c_j) = 2 - 2s \leq 3. \end{aligned}$$

If equality holds, then $s = -1/2$ and equality in arithmetic-geometric mean forces all $c_j = -1/2$. Every orbit point is therefore $1/3$ or $2/3$ modulo one, so the reduced orbit is exactly $\{1/3, 2/3\}$ and $q = 3$. The converse is immediate. $\square$

Corollary 6.2 (Sharp minimum of the rational-ray exponent). *Every nonzero rational ray satisfies*

$$\kappa_{a,q} \geq \kappa_{\min} := \frac{1}{2} + \frac{\log(2\pi/\sqrt{3})}{\log 2} = 2.3590148791117407073 \dots \quad (40)$$

Equality is attained only by the denominator-three doubling orbit.

NEW RESULT PROVED HERE.

Proof. Insert (39) into (20). $\square$

Remark 6.3 (A spectral interpretation). The arithmetic exponents form a spectrum of cycle averages of $\log |2 \sin(\pi t)|$. The theorem identifies the exact lower endpoint without using transfer-operator machinery. Section 7 constructs exponents tending to $+\infty$, so the spectrum has a sharp bottom and no finite top.

7 Mersenne and plus-denominator rays

The preceding theory turns a fast-decay construction into an arithmetic design problem: choose a rational doubling cycle for which $A_{a,q}$ is very small. The simplest such cycles come from the denominators immediately adjacent to a power of two.

7.1 Exact orbit products

For $d \geq 2$ put

$$q_d^- := 2^d - 1, \quad q_d^+ := 2^d + 1, \quad (41)$$

and let A_d^- and A_d^+ denote the amplitudes $A_{1,q}$ for these denominators.

Lemma 7.1 (Orders of two). *One has*

$$\text{ord}_{q_d^-}(2) = d, \quad \text{ord}_{q_d^+}(2) = 2d. \quad (42)$$

Proof. For q_d^- , if $2^r \equiv 1 \pmod{2^d - 1}$ with $0 < r < d$, then the positive integer $2^r - 1$ would be divisible by the larger integer $2^d - 1$, impossible.

For q_d^+ , $2^d \equiv -1$, so the order r is even. Inside the cyclic group generated by 2, the unique element of order two is $2^{r/2}$; hence $d = (2m + 1)r/2$ for some $m \geq 0$. If $m \geq 1$, then $r \leq 2d/3 < d$, but q_d^+ would divide $2^r - 1 < 2^d + 1$, again impossible. Thus $m = 0$ and $r = 2d$. $\square$

Proposition 7.2 (Exact product forms). *For $d \geq 2$,*

$$A_d^- = \prod_{k=1}^d 2 \sin\left(\frac{\pi 2^{-k}}{1 - 2^{-d}}\right), \quad (43)$$

$$A_d^+ = \left[\prod_{k=1}^d 2 \sin\left(\frac{\pi 2^{-k}}{1 + 2^{-d}}\right) \right]^2. \quad (44)$$

Proof. For q_d^- , the orbit is $1, 2, \dots, 2^{d-1}$ under multiplication by two. Reindexing $j = d - k$ gives

$$\frac{2^j}{2^d - 1} = \frac{2^{-k}}{1 - 2^{-d}}.$$

For q_d^+ the order is $2d$ and $2^{j+d} \equiv -2^j \pmod{q_d^+}$, so the second half of the orbit contributes the same absolute factors as the first. The same reindexing proves (44). $\square$

7.2 The universal sinc-tail constant

Define

$$B_d := \prod_{k=1}^d 2 \sin\left(\frac{\pi}{2^k}\right), \quad H_d(1) := \prod_{k=1}^d \operatorname{sinc}\left(\frac{\pi}{2^k}\right). \quad (45)$$

Then

$$B_d = (2\pi)^d 2^{-d(d+1)/2} H_d(1). \quad (46)$$

The finite sinc tail converges to

$$h := \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{\pi}{2^k}\right) = 0.5537712758888100953442174163445488813\dots \quad (47)$$

Lemma 7.3 (Three forms of the tail constant). *The constant h satisfies*

$$h = -\mathcal{U}'(1), \quad (48)$$

$$\log h = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(4^m - 1)}. \quad (49)$$

Moreover,

$$\log H_d(1) = \log h + \mathcal{O}(4^{-d}). \quad (50)$$

Proof. From (2),

$$\mathcal{U}(x) = \operatorname{sinc}(\pi x) \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{2^k}\right).$$

The derivative of $\operatorname{sinc}(\pi x)$ at $x = 1$ is -1 , proving (48). Euler's product for the sine gives, for $|z| < \pi$,

$$\log \operatorname{sinc} z = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}} z^{2m}.$$

Substitute $z = \pi/2^k$, sum over $k \geq 1$, and use $\sum_{k \geq 1} 4^{-km} = 1/(4^m - 1)$. Finally, $\log \operatorname{sinc}(\pi/2^k) = \mathcal{O}(4^{-k})$, which yields (50). $\square$

The first exponentially small correction is controlled by the convergent constant

$$C_{\cot} := \sum_{k=1}^{\infty} \left(\pi 2^{-k} \cot(\pi 2^{-k}) - 1 \right) = -1.2837124730461260445945438959726042474 \dots \quad (51)$$

Convergence follows from $x \cot x = 1 - x^2/3 + \mathcal{O}(x^4)$ at the origin.

Theorem 7.4 (Refined twin product expansion). *As $d \rightarrow \infty$,*

$$\log \frac{A_d^-}{B_d} = (d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d}), \quad (52)$$

$$\log \frac{\sqrt{A_d^+}}{B_d} = -(d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d}). \quad (53)$$

Equivalently,

$$A_d^- = h(2\pi)^d 2^{-d(d+1)/2} \exp\left((d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d})\right), \quad (54)$$

$$A_d^+ = h^2(2\pi)^{2d} 2^{-d(d+1)} \exp\left(-2(d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d})\right). \quad (55)$$

NEW RESULT PROVED HERE.

Proof. Put $\delta = 2^{-d}$, $x_k = \pi 2^{-k}$, and $g(x) = \log(2 \sin x)$. From [theorem 7.2](#),

$$\begin{aligned} \log(A_d^-/B_d) &= \sum_{k=1}^d \left[g\left(\frac{x_k}{1-\delta}\right) - g(x_k) \right], \\ \log(\sqrt{A_d^+}/B_d) &= \sum_{k=1}^d \left[g\left(\frac{x_k}{1+\delta}\right) - g(x_k) \right]. \end{aligned}$$

On the relevant interval, $xg'(x) = x \cot x$ and $x^2 g''(x)$ are uniformly bounded. Taylor expansion under a multiplicative perturbation therefore gives, uniformly in k ,

$$g\left(\frac{x_k}{1 \mp \delta}\right) - g(x_k) = \pm \delta x_k \cot x_k + \mathcal{O}(\delta^2).$$

Summing produces an error $\mathcal{O}(d\delta^2)$. Further,

$$\sum_{k=1}^d x_k \cot x_k = d + C_{\cot} + \mathcal{O}(4^{-d}).$$

This proves (52)–(53). Equations (54)–(55) follow from (46) and (50). $\square$

7.3 Exponent asymptotics and unboundedness

Let κ_d^- and κ_d^+ be the ray exponents for q_d^- and q_d^+ , respectively, and put

$$K_d := \frac{d}{2} + 1 - \frac{\log h}{d \log 2}. \quad (56)$$

Corollary 7.5 (Opposite first corrections). *As $d \rightarrow \infty$,*

$$\kappa_d^- = K_d - \frac{2^{-d}}{\log 2} \left(1 + \frac{C_{\cot}}{d} \right) + \mathcal{O}(4^{-d}), \quad (57)$$

$$\kappa_d^+ = K_d + \frac{2^{-d}}{\log 2} \left(1 + \frac{C_{\cot}}{d} \right) + \mathcal{O}(4^{-d}). \quad (58)$$

In particular,

$$\kappa_d^\pm = \frac{d}{2} + 1 - \frac{\log h}{d \log 2} + \mathcal{O}(2^{-d}), \quad (59)$$

and the rational-ray exponent spectrum is unbounded above.

NEW RESULT PROVED HERE.

Proof. Insert (54) into (20) with order d , and (55) with order $2d$. The terms $\log(2\pi)$ cancel against κ_0 , and the remaining algebra gives (57) and (58). Since $K_d \sim d/2$, the exponents tend to infinity. $\square$

Combining theorems 6.2 and 7.5 gives a complete endpoint statement.

Corollary 7.6 (Endpoint geometry of the rational spectrum). *Let $\mathcal{K}_{\text{rat}}$ be the set of exponents of all nonzero rational rays. Then*

$$\min \mathcal{K}_{\text{rat}} = \kappa_{\min}, \quad \sup \mathcal{K}_{\text{rat}} = +\infty. \quad (60)$$

The minimum is attained only by denominator 3.

8 A direct half-base q -binomial bridge

The Mersenne factorial (7) is the denominator-clearing normalization behind the half-base Gaussian coefficients. The preceding product asymptotics show that the same arithmetic factors normalize special cyclotomic ray multipliers.

Set

$$\mathcal{C}_* := (1/2; 1/2)_\infty h = 0.1599225518776068325779014528557986690 \dots, \quad (61)$$

where

$$(1/2; 1/2)_\infty = 0.2887880950866024212788997219292307801 \dots \quad (62)$$

Theorem 8.1 (Mersenne–cyclotomic–sinc bridge). *As $d \rightarrow \infty$,*

$$\boxed{\frac{\mathfrak{M}_d A_d^-}{(2\pi)^d} \longrightarrow \mathcal{C}_*}, \quad \boxed{\frac{\mathfrak{M}_d \sqrt{A_d^+}}{(2\pi)^d} \longrightarrow \mathcal{C}_*}. \quad (63)$$

More precisely,

$$\frac{\mathfrak{M}_d A_d^-}{(2\pi)^d} = \mathcal{C}_* \exp\left((d + C_{\cot} + 1)2^{-d} + \mathcal{O}(d4^{-d})\right), \quad (64)$$

$$\frac{\mathfrak{M}_d \sqrt{A_d^+}}{(2\pi)^d} = \mathcal{C}_* \exp\left((1 - d - C_{\cot})2^{-d} + \mathcal{O}(d4^{-d})\right). \quad (65)$$

NEW RESULT PROVED HERE.

Proof. Using (8) and (46),

$$\frac{\mathfrak{M}_d B_d}{(2\pi)^d} = (1/2; 1/2)_d H_d(1).$$

Now multiply by A_d^-/B_d or by $\sqrt{A_d^+}/B_d$ and use [theorem 7.4](#). The limits follow because $(1/2; 1/2)_d \rightarrow (1/2; 1/2)_\infty$ and $H_d(1) \rightarrow h$.

For the refined forms, note

$$\log \frac{(1/2; 1/2)_d}{(1/2; 1/2)_\infty} = - \sum_{k>d} \log(1 - 2^{-k}) = 2^{-d} + \mathcal{O}(4^{-d}),$$

while (50) contributes only $\mathcal{O}(4^{-d})$. Adding the exponent corrections from (52) and (53) proves (64)–(65). $\square$

Interpretation. The limit (63) is a literal bridge among the three recurrent structures of the repository:

$$\underbrace{\mathfrak{M}_d}_{\text{half-base Gaussian normalization}} \times \underbrace{A_d^- \text{ or } \sqrt{A_d^+}}_{\text{cyclotomic Thue–Morse orbit norm}} \times \underbrace{(2\pi)^{-d}}_{\text{sinc scaling}} \longrightarrow \underbrace{(1/2; 1/2)_\infty (-\mathcal{U}'(1))}_{\text{global constant}}.$$

This is more than a shared appearance of powers of two: it is a quantitative asymptotic identity tying the q -binomial and Fourier sides together.

8.1 Numerical convergence of the twin exponents

d	$2^d - 1$	κ_d^-	$2^d + 1$	κ_d^+	K_d
2	3	2.35901487911	5	2.57101410575	2.42631893591
3	7	2.68360364246	9	2.88733571269	2.78421262394
4	15	3.15149612947	17	3.27378362740	3.21315946796
5	31	3.63677701126	33	3.70378409640	3.67052757436
6	63	4.12430649454	65	4.15974429455	4.14210631197
7	127	4.61257768880	129	4.63098586366	4.62180541026
8	255	5.10184195614	257	5.11130441973	5.10657973398
9	511	5.59231992724	513	5.59715163452	5.59473754131
10	1023	6.08403530020	1025	6.08649134438	6.08526378718
11	2047	6.57689018061	2049	6.57813464441	6.57751253380
12	4095	7.07073858328	4097	7.07136766593	7.07105315599
13	8191	7.56542880068	8193	7.56574624039	7.56558752860
14	16383	8.06082272205	16385	8.06098268408	8.06090270513

Table 2: Mersenne and plus-denominator exponents bracketing the common main term K_d . The gap has the predicted size $2^{1-d}/\log 2$ to first order.

9 Exact computations and orbit-polynomial data

The theory is computationally concrete. For a given odd q one needs only the multiplicative order of 2, a finite product at roots of unity, and either a resultant or Ramanujan power sums. No numerical integration of up is involved.

9.1 Two exact algorithms

Algorithm 9.1 (Resultant route). Given odd $q > 1$:

1. compute $d = \text{ord}_q(2)$;
2. form $F_d(z) = \prod_{j=0}^{d-1} (1 - z^{2^j})$;
3. compute $R(X) = \text{Res}_z(\Phi_q(z), X - F_d(z))$ over $\mathbb{Z}[X]$;
4. factor R and divide every factor exponent by d ;
5. the resulting exact polynomial is $\mathcal{P}_{q,2}(X)$;
6. take the absolute values of its complex roots and insert them into (20).

The perfect- d th-power assertion in (30) is a built-in check.

Algorithm 9.2 (Ramanujan–Newton route). Let $c_q^* = \varphi(q)/d$. Compute the power sums $p_r = \sum_C \Gamma_C^r$ for $1 \leq r \leq c_q^*$ by (32), then apply Newton’s identities to recover the elementary symmetric functions and hence $\mathcal{P}_{q,2}(X)$. This route uses integer arithmetic and can exploit fast convolution of the finite Thue–Morse block.

The resultant route is simplest for modest d . The Ramanujan route becomes attractive when 2^d is too large to expand naively but q has a simple divisor structure, because $c_q(m)$ depends only on (q, m) .

9.2 Exact orbit polynomials

q	$d = \text{ord}_q(2)$	$\mathcal{P}_{q,2}(X)$
3	2	$X - 3$
5	4	$X - 5$
7	3	$X^2 + 7$
9	6	$X - 3$
11	10	$X - 11$
15	4	$(X + 1)^2$
17	8	$X^2 - 34X + 17$
21	6	$(X + 1)^2$
23	11	$X^2 + 23$
31	5	$X^6 + 93X^4 + 899X^2 + 31$

Table 3: Exact orbit polynomials obtained by the resultant method. Repeated roots at $q = 15$ and 21 reflect a larger stabilizer than $\langle 2 \rangle$.

Several structural features are visible immediately.

- If 2 generates all of G_q , the orbit polynomial is linear and the multiplier is $\Phi_q(1)$; this explains $q = 3, 5, 9, 11$.
- For $q = 7$ and 23, the two multipliers are purely imaginary conjugates and both amplitudes equal $\sqrt{q}$.
- At $q = 17$, the two multipliers are real and split as $17 \pm 4\sqrt{17}$.
- For $q = 31$, the odd order $d = 5$ makes the multiplier roots occur in imaginary pairs; the squared amplitudes are the positive roots of $Y^3 - 93Y^2 + 899Y - 31$.

q	multiplicity	amplitude $A_{a,q}$	exponent $\kappa_{a,q}$
3	1	3	2.3590148791117
5	1	5	2.5710141057505
7	2	$\sqrt{7} = 2.6457513110646\dots$	2.6836036424627
9	1	3	2.8873357126855
11	1	11	2.8055529676086
15	2	1	3.1514961294723
17	1	$17 - 4\sqrt{17} = 0.50757749752936\dots$	3.2737836273969
17	1	$17 + 4\sqrt{17} = 33.492422502471\dots$	2.5182757763915
21	2	1	3.1514961294723
23	2	$\sqrt{23} = 4.7958315233127\dots$	2.9458796769243
31	2	0.18602850302165...	3.6367770112610
31	2	3.3042170276289...	2.8066344396753
31	2	9.0580098934805...	2.5156573064420

Table 4: Amplitude classes and rational-ray exponents. Multiplicity is the number of H_q -cosets with that amplitude.

9.3 Exact checks performed

The calculations behind tables 3 and 4 were checked in three independent ways:

1. the symbolic resultant was factored and verified to be a perfect d th power;
2. high-precision direct products $\prod_{j < d} (1 - \zeta_q^{a2^j})$ were evaluated for every coset and substituted into the exact integer polynomial;
3. the product of all orbit amplitudes was checked against $\Phi_q(1)$, and the numerical mean of the exponents was checked against (37).

The fixed-mantissa factorization (11) was also evaluated at $y = 4/3, 6/5, 8/7$ and $N = 0, 1, 3, 10$ with eighty decimal digits. The largest relative discrepancy was below 3×10^{-78} .

9.4 Reproducible symbolic code

The following compact SymPy program implements the resultant route. It is included as mathematical pseudocode rather than as a software dependency of the document.

Listing 1: Orbit-polynomial computation.

```
import sympy as sp
from math import gcd

def order_mod_2(q: int) -> int:
    x = 1
    for d in range(1, 2*q + 1):
        x = (2*x) % q
        if x == 1:
            return d
    raise ArithmeticError("order not found")

def orbit_polynomial(q: int):
    if q <= 1 or q % 2 == 0:
        raise ValueError("q must be odd and > 1")
    z, X = sp.symbols("z X")
    d = order_mod_2(q)
    F = sp.prod(1 - z**(2**j) for j in range(d))
    Phi = sp.cyclotomic_poly(q, z)
    R = sp.Poly(sp.resultant(Phi, X - F, z), X)
    factors = sp.factor_list(R.as_expr(), X)[1]
```

```

P = sp.Integer(1)
for factor, exponent in factors:
    if exponent % d:
        raise ArithmeticError("resultant is not a d-th power")
    P *= factor**(exponent // d)
return d, sp.Poly(sp.expand(P), X)

for q in [3, 5, 7, 9, 11, 15, 17, 21, 23, 31]:
    d, P = orbit_polynomial(q)
    print(q, d, sp.factor(P.as_expr()))

```

10 A general-base extension

The arithmetic mechanism does not depend on base two until the sharp endpoint argument. For an integer $b \geq 2$, define

$$\mathcal{U}_b(x) := \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{b^n}\right). \quad (66)$$

For $1 < y < 2$ and $t = y - 1$, put

$$H_b(y) := \prod_{r=1}^{\infty} \operatorname{sinc}\left(\frac{\pi y}{b^r}\right), \quad Q_{b,n}(t) := \prod_{j=0}^{n-1} |2 \sin(\pi b^j t)|. \quad (67)$$

(The interval $1 < y < 2$ is enough for a canonical family of rays; one may use $1 < y < b$ with only notational changes.)

Theorem 10.1 (Exact rational rays in base b). *For every $N \geq 0$,*

$$|\mathcal{U}_b(b^N y)| = b^{-N(N+1)/2} (\pi y)^{-(N+1)} H_b(y) 2^{-(N+1)} Q_{b,N+1}(t). \quad (68)$$

Let $q > 1$, $(a, q) = 1$, $(b, q) = 1$, $d = \operatorname{ord}_q(b)$, and

$$A_{a,q}^{(b)} := \left| \prod_{j=0}^{d-1} (1 - \zeta_q^{ab^j}) \right|. \quad (69)$$

On the ray $x_N = b^N(1 + a/q)$ there is an exact periodic law

$$\log |\mathcal{U}_b(x_N)| = -\frac{(\log x_N)^2}{2 \log b} - \kappa_{a,q}^{(b)} \log x_N + \Omega_{a,q}^{(b)}(N), \quad (70)$$

where $\Omega_{a,q}^{(b)}$ has period dividing d and

$$\kappa_{a,q}^{(b)} = \frac{1}{2} + \frac{\log(2\pi)}{\log b} - \frac{\log A_{a,q}^{(b)}}{d \log b}. \quad (71)$$

If $H_{q,b} = \langle b \rangle \subseteq (\mathbb{Z}/q\mathbb{Z})^\times$, then the mean over its cosets is

$$\frac{1}{|G_q/H_{q,b}|} \sum_C \kappa_C^{(b)} = \frac{1}{2} + \frac{\log(2\pi)}{\log b} - \frac{\log \Phi_q(1)}{\varphi(q) \log b}. \quad (72)$$

NEW RESULT PROVED HERE.

Proof. Split the product at $n = N$ exactly as in [theorem 3.1](#); the denominator now contributes $b^{N(N+1)/2}$, while the factor $2^{-(N+1)}$ still comes from replacing $|\sin|$ by $|2\sin|$. If $t = a/q$, multiplication by b has period d modulo q , so $Q_{b,n+d} = A_{a,q}^{(b)} Q_{b,n}$. Completing the square in $\log x_N = N \log b + \log y$ gives (71). The mean law follows from the same norm identity because the $H_{q,b}$ -cosets partition G_q . $\square$

Remark 10.2 (What is genuinely binary). The resultant, relative norm, exact periodic remainder, and mean-exponent law are base-independent. The sharp $\sqrt{3}$ endpoint uses the special identity $\cos(2x) = 2\cos^2 x - 1$ and the fact that squaring permutes the cycle. Finding the corresponding sharp constant for $b \geq 3$ is a distinct ergodic-optimization problem.

11 A Lean formalization blueprint

The new results are deliberately organized so that most proof obligations are finite algebra, elementary real analysis, or existing cyclotomic infrastructure. This section gives a concrete path from the report to the repository’s formalized core.

11.1 Suggested module boundary

`Analysis/FabiusFunction/ThueMorse/CyclotomicOrbit.lean`

Define T_n , prove its product formula, evaluate it on a root of unity, and prove the block recurrence $T_{n+d} = \Gamma T_n$.

`Analysis/FabiusFunction/Fourier/RationalRay.lean`

Split the infinite sinc product at N , prove (11), and derive the exact periodic log law.

`Analysis/FabiusFunction/NumberTheory/OrbitPolynomial.lean`

Define $H_q = \langle 2 \rangle$, the relative norm, and the orbit polynomial. Prove integrality, the norm product, and the mean exponent identity.

`Analysis/FabiusFunction/Analysis/MersenneRay.lean`

Establish the exact products (43)–(44), the convergence to h , and then the asymptotic estimates.

`Analysis/FabiusFunction/QCalculus/RayBridge.lean`

Combine the Mersenne factorial identity with the sinc-tail limit to formalize (63).

11.2 Core theorem signatures

The exact identities can be formalized before any logarithms, avoiding extended real values at zeros. Indicative signatures are:

Listing 2: Indicative Lean theorem interfaces.

```
-- Finite Thue-Morse polynomial at a root of unity.
theorem tmPoly_eval_cycle
  (hq : Odd q) (ha : Nat.Coprime a q) :
  tmPoly (n + orderOfTwo q) (zeta q ^ a) =
    cycleMultiplier a q * tmPoly n (zeta q ^ a)

-- Exact absolute factorization of the Rvachev Fourier product.
theorem upFourier_abs_dyadic_mantissa
  (hy : 1 < y) (hy2 : y < 2) :
  abs (upFourier ((2 : R) ^ N * y)) =
    2 ^ (-(N * (N + 1) / 2 : Z)) *
    (Real.pi * y) ^ (-(N + 1 : Z)) *
    sincTail y * 2 ^ (-(N + 1 : Z)) *
```

```

orbitSineProduct (N + 1) (y - 1)

-- Product over G/H and the cyclotomic value at one.
theorem prod_cycleMultiplier_cosets :
  Finset.prod (allCosets q) (fun C =>
    cycleMultiplier C.rep q) = cyclotomic q 1

-- Sharp periodic-orbit bound.
theorem cycleAmplitude_geomMean_le_sqrt_three :
  Real.rpow (cycleAmplitude a q) (1 / orderOfTwo q) <= Real.sqrt 3

```

The displayed syntax is schematic. The main implementation decision is whether to represent roots of unity in $\mathbb{C}$ or in an abstract cyclotomic extension. The resultant theorem is easiest abstractly; the absolute-value ray law is easiest in $\mathbb{C}$ with the standard embedding.

11.3 Proof dependency order

A low-risk formalization sequence is:

1. finite product and block recurrence;
2. exact fixed-mantissa factorization without logarithms;
3. rational preperiod/period dichotomy;
4. norm product and mean exponent;
5. sharp $\sqrt{3}$ inequality, which uses only finite real sums and AM–GM;
6. analytic Mersenne estimates;
7. resultant and Ramanujan trace formulas, which can be added independently.

The first five steps already formalize the central qualitative discoveries and do not depend on delicate asymptotic libraries.

12 Further frontier problems

The exact arithmetic-ray theory raises a set of focused questions that are sharper than the generic request to “understand Fourier decay.”

12.1 The closure of the ray-exponent spectrum

Conjecture 12.1 (Interval closure). *Let*

$$\mathcal{K}_{\text{rat}} = \{\kappa_{a,q} : q > 1 \text{ odd}, (a, q) = 1\}.$$

Its closure may be the whole interval $[\kappa_{\min}, \infty)$.

OPEN OR CONJECTURAL.

Periodic orbit measures are dense among invariant measures of the doubling map for continuous observables, but $\log |2 \sin(\pi t)|$ has logarithmic singularities. The conjecture therefore requires a controlled specification argument that keeps periodic words away from zero while prescribing their cycle average. The Mersenne family gives the unbounded direction, and the denominator-three orbit gives the lower endpoint; the missing issue is interpolation.

12.2 Extremizers at fixed period

For each d , determine

$$A_d^{\max} := \max_{\text{ord}_q(2)=d} A_{a,q}, \quad A_d^{\min} := \min_{\text{ord}_q(2)=d} A_{a,q}.$$

The sharp global maximum of $A^{1/d}$ is known from [theorem 6.1](#), but the finite-period problem may have a rich symbolic structure. A successful solution would give the sharpest and slowest rational rays at every scale period.

12.3 Stabilizers and repeated orbit-polynomial roots

Characterize the subgroup

$$\text{Stab}(\Gamma_{1,q}) = \{u \in G_q : \Gamma_{u,q} = \Gamma_{1,q}\}.$$

It contains H_q but can be larger, as $q = 15$ and 21 show. Equivalently, determine when $\mathcal{P}_{q,2}$ is squarefree and when it is a proper power. This is a concrete problem in cyclotomic units and automatic polynomial values.

12.4 Regulator statistics

When q has at least two prime divisors, [\(36\)](#) says $\sum_C \log A_C = 0$. The vector $(\log A_C)_C$ lies in the logarithmic embedding of a cyclotomic-unit group. Its covariance, extreme coordinates, and limiting distribution as $q \rightarrow \infty$ should connect ray decay with regulators and unit lattices. Prime-power denominators add the deterministic shift $\log p/\varphi(p^s)$ from [\(37\)](#).

12.5 Beyond the first Mersenne correction

The proof of [theorem 7.4](#) can be iterated. Taylor expansion in $\delta = 2^{-d}$ produces an all-orders transseries whose coefficients are convergent cotangent/cosecant sums over dyadic angles, together with the tail expansion of $H_d(1)$. Determining a closed algebra for these coefficients would parallel the periodic saddle-jet machinery already developed for the small-argument Fabius asymptotic.

12.6 A wider q -binomial bridge

The bridge theorem uses the terminal Mersenne factor $2^d - 1$ and the full product $\mathfrak{M}_d = \prod_{j \leq d} (2^j - 1)$. A broader goal is to connect general cyclotomic orbit norms $A_{a,q}$ to coefficient extraction in the half-base q -binomial formulas. Possible entry points are:

- denominators dividing products of Mersenne integers;
- discrete Fourier transforms of the Gaussian interpolation blocks at roots of unity;
- resultant identities in which $F_d(z)$ is paired with the half-base Pochhammer polynomial.

A direct finite identity here would join the exact dyadic values of F to the arithmetic rays of $\widehat{\text{up}}$, rather than linking them only asymptotically.

12.7 Sharp constants in other bases

For $b \geq 3$, determine

$$\sup_{q,a} \left(A_{a,q}^{(b)} \right)^{1/\text{ord}_q(b)}$$

and classify the maximizing periodic orbits. The binary proof is exceptional because the real part of $z \mapsto z^2$ closes quadratically. The general problem looks like a finite-dimensional ergodic-optimization problem for $\log |1 - z|$ under $z \mapsto z^b$.

13 Conclusion

The reviewed corpus already contains two powerful but largely parallel bodies of work: exact $q = 1/2$ arithmetic for Fabius values and dynamical Fourier decay for the Rvachev product. The arithmetic-ray viewpoint inserts the finite Thue–Morse polynomial between them. On rational mantissas, the doubling cocycle closes; its cycle multiplier is a relative cyclotomic norm, and the Fourier decay becomes an exact log-quadratic formula with a finite periodic remainder.

This perspective yields several rigid structures at once:

- an integral orbit polynomial and a perfect-power resultant;
- Ramanujan-sum formulas for every multiplier trace;
- a one-integer mean-exponent law governed by $\Phi_q(1)$;
- a sharp rational-ray minimum from the orbit $\{1/3, 2/3\}$;
- unbounded exponents from Mersenne denominators;
- a twin Mersenne/plus-denominator expansion with opposite first corrections;
- a direct limit identity joining cyclotomic norms, the half-base q -Pochhammer constant, and $-\mathcal{U}'(1)$.

The results are elementary enough to be formalized in stages, yet arithmetic enough to open new questions about cyclotomic units, regulators, periodic-orbit spectra, and Gaussian q -binomial transforms. They therefore fit the repository’s intended frontier role: mathematically testable statements, clearly separated from imported results, and organized for eventual migration into Lean.

A Corpus inventory

The audit counted fifty-seven TeX files: twenty-five in the living documentation tree and thirty-two frozen historical versions. The list below records the path relative to `Analysis/FabiusFunction/docs`. Historical items are grouped by the archive directory; filenames follow the base name shown.

A.1 Living documentation: 25 TeX files

No.	Relative path
1	<code>FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex</code>
2	<code>Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex</code>
3	<code>Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex</code>
4	<code>fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>
5	<code>non-formalized-research-frontiers/Non_Formalized_Research_Frontiers.tex</code>
6	<code>non-formalized-research-frontiers/drafts/Fabius_Thue_Morse_Formula_Atlas/Fabius_Thue_Morse_Formula_Atlas.tex</code>
7	<code>non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex</code>
8	<code>non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex</code>

No.	Relative path
9	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
16	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
18	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
19	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
20	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
21	papers/arXiv-1502.06738v2/Lacunary_products.tex
22	papers/arXiv-1702.05442v1/09-Function.tex
23	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24	papers/arXiv-2211.13570v1/document.tex
25	papers/ubiq15/ubiq15-corrected.tex

A.2 Frozen archive: 32 TeX files

No.	Relative path under archive/
1	early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
2	early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
3	standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
5	standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

No.	Relative path under archive/
6	primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
7	primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
8	primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
9	primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
10	primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
11	primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
12	primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
13	primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
14	q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
15	q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
16	lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
17	lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
18	glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
19	glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
20	research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
21	research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
22	research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
23	research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
24	research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
25	research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
26	research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
27	research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
28	research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
29	research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex

No.	Relative path under archive/
30	research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
31	research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
32	research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Constants and numerical reference

For convenient independent checking:

$$\begin{aligned}
\kappa_0 &= 3.1514961294723187980432792951080073350\dots, \\
\kappa_{\min} &= 2.3590148791117407073164098231340990806\dots, \\
h &= 0.5537712758888100953442174163445488813\dots, \\
-\log_2 h &= 0.8526378718208337174366143795551934080\dots, \\
C_{\cot} &= -1.2837124730461260445945438959726042474\dots, \\
(1/2; 1/2)_{\infty} &= 0.2887880950866024212788997219292307801\dots, \\
C_{*} &= 0.1599225518776068325779014528557986690\dots
\end{aligned}$$

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